

A Number-Valued Broadcast Sum of Surreal Sequences

Commuting truncations and an exact failure of monotonicity

A partial attack on Lipparini's Problem 7.5

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Abstract. We analyze the sign-truncation rule proposed by Lipparini for extending an infinitary ordinal operation to surreal sequences. A player chooses a sign position and applies the associated truncation to all but finitely many coordinates. We supply an explicit descending ordinal rank and prove that every resulting game is a number: opposite truncations commute, and the earlier selected sign survives the other move. The operation agrees with ordinary surreal addition on finite-support families and with Lipparini's operation on ordinal sequences. A finite-birthday perturbation bound permits exact calculations for cofinite dyadic families. The main example is a positive, classically convergent series with terms either $2^{-(n+d)}$ or $3 \cdot 2^{-(n+d+1)}$. Its broadcast value is $k + 1$ if exactly k terms take the second value, and $\sqrt{\omega}$ if infinitely many do. A strictly larger geometric sequence has broadcast value 1. Thus the proposed rule is number-valued but fails even weak coordinatewise monotonicity, including on absolutely convergent positive dyadic sequences. Full proofs and an exact finite verification program are provided.

Status and scope. The underlying rule is Lipparini's, not a new definition claimed here. The proofs below are research claims developed in this manuscript; they have not been independently refereed or formally verified. The result analyzes a published candidate for an open-ended extension problem. It does *not* construct a weakly monotone surreal summation theory or prove that no such alternative is possible. A limited literature check does not establish priority.

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Reading guide. Sections 1–3 state the target and the rule. Sections 4–7 establish the structural theorems. Sections 8–10 contain the exact dyadic calculation and counterexamples. Sections 11–13 and the appendices document verification, limitations, and reproducibility.

1 The target and the outcome

Lipparini’s extended paper [5] asks in Problem 7.5 for a suitable surreal or game-theoretic extension of the ordinal game in Definition 7.1. Remark 7.6(3) proposes the sign-truncation rule studied here and already observes that its runs terminate. The version checked is arXiv:2505.00424v2, dated April 30, 2026; the relevant discussion is on printed pages 36–37. We use the local name *broadcast sum* and the notation $\mathbf{B}$ for this rule.

There are two distinct questions. Does the rule yield a surreal number rather than an arbitrary game? And does it retain useful order or summation properties? Our answer is positive to the first question and sharply negative to the second.

1.1 Main results in a compact form

For every set-indexed family $\mathbf{s} = (s_i)_{i \in I}$ of canonical surreal sign expansions, the rule defines a well-founded game $G(\mathbf{s})$. We prove

$$G(\mathbf{s}) \text{ is a number,} \quad \mathbf{B}(\mathbf{s}) \in \mathbf{No}. \quad (1)$$

The passage from countable to set-indexed families uses the same finite-exemption rule; no cardinal-dependent normalization is introduced.

For a concrete exact calculation, let $d \geq 2$ be an integer and $E \subseteq \mathbb{N}$, where $\mathbb{N} = \{0, 1, 2, \dots\}$. Put

$$s_n^{E,d} = \begin{cases} 3 \cdot 2^{-(n+d+1)}, & n \in E, \\ 2^{-(n+d)}, & n \notin E. \end{cases} \quad (2)$$

Then

$$\mathbf{B}((s_n^{E,d})_{n < \omega}) = \begin{cases} |E| + 1, & E \text{ finite,} \\ \sqrt{\omega}, & E \text{ infinite.} \end{cases} \quad (3)$$

Here $\sqrt{\omega}$ is the positive square root in the surreal field. It is not ordinal exponentiation by a fractional exponent. In Conway monomial notation it is $\omega^{1/2}$.

Every series in (2) is an ordinary positive real series, and

$$0 < \sum_{n=0}^{\infty} s_n^{E,d} \leq 3 \cdot 2^{-d} < 1. \quad (4)$$

Nevertheless, the broadcast output can exceed every real number. More strongly, $s_n^{E,d} < 2^{-(n+d-1)}$ for every n , whereas

$$\mathbf{B}((2^{-(n+d-1)})_{n < \omega}) = 1. \quad (5)$$

Thus (3) is an explicit counterexample to weak monotonicity whenever $E \neq \emptyset$.

1.2 What is and is not resolved

The manuscript supplies a numerical extension of the ordinal rule and a substantial analysis of that extension. It also rules out an attractive property of this particular candidate. This is a partial resolution of the stated research direction, not a claim that the word “suitable” has acquired a unique mathematical meaning.

The strongest new claims proposed here are the general number-valuedness theorem, the finite-short-perturbation estimate, and the exact classification (3). The explicit rank is a quantitative proof of termination, not a claim to originate the termination observation. Standard background on numbers, cuts, and game comparison is used throughout [1, 2, 6]. Other infinitary game constructions, such as [3, 4], should not be identified with the rule analyzed here.

2 Conventions and elementary surreal background

2.1 Signs, prefixes, and options

We work in ZFC. Index families are set-sized; **No** is the proper class of surreal numbers, used through set-sized sign expansions and cuts.

A surreal number has a unique canonical sign expansion $s : \ell(s) \rightarrow \{+, -\}$, with ordinal length $\ell(s)$. Its canonical left options are prefixes immediately preceding a plus sign; its canonical right options are prefixes immediately preceding a minus sign:

$$s = \{ s \restriction \gamma : s(\gamma) = + \mid s \restriction \gamma : s(\gamma) = - \}. \quad (6)$$

Only positions $\gamma < \ell(s)$ occur here. A prefix preceding a plus is strictly smaller than s , and a prefix preceding a minus is strictly larger. The empty string is 0. The all-plus string of length α represents the ordinal α .

A *short* number in this paper means a number of finite birthday, or equivalently a dyadic rational represented by a finite sign string. In particular, “short” does not mean every real number. A *finite* surreal means one whose absolute value is bounded by some natural number; such a number need not be real or dyadic.

We use game forms when discussing options and their numerical equivalence classes when calculating values. A game is recursively a *number* when its options are numbers and every left option is strictly smaller than every right option. This recursive requirement matters: separation of options without their number-valuedness is not enough.

2.2 Comparison and cofinal replacement of cuts

For numerical game forms x and y , the usual comparison test is

$$x \leq y \iff (\forall x^L x^L < y) \text{ and } (\forall y^R x < y^R). \quad (7)$$

A separated cut denotes the simplest surreal strictly between its two sides. Replacing a left side by a mutually cofinal set, or a right side by a mutually cointial set, leaves the value unchanged. For example, any unbounded set of nonnegative real numbers is mutually cofinal with $\mathbb{N}$.

When L and R are sets of numbers, “mutually cointial” means that each member of either set lies above or equals a member of the other. This is an order statement, not the assertion that either set has an infimum in **No**. These basic tools follow from the ordinary theory of numbers and the simplicity theorem; see, for instance, [6, Sections 3–5].

2.3 Arithmetic conventions

We write $\alpha \# \beta$ for the finite Hessenberg natural sum of ordinals. In common Cantor normal form it adds the corresponding finite coefficients; it is strictly increasing in either argument. The natural

sum of a finite family includes every indexed occurrence, with multiplicity.

In Section 4, addition, multiplication, and powers of ordinals without $\#$ mean *ordinary ordinal* operations. In the remaining sections, products such as $\omega 2^{-m}$ and expressions such as $\omega - k$ mean *surreal field* arithmetic. This distinction is essential: $\omega/2$ and $\omega - 1$ are surreals, not ordinals.

For later use, the standard Conway omega-map satisfies

$$\omega^x = \left\{ 0, n\omega^{x^L} : n \in \mathbb{N}_{>0} \mid 2^{-n}\omega^{x^R} : n \in \mathbb{N} \right\}, \quad \omega^{x+y} = \omega^x \omega^y. \quad (8)$$

Using $1/2 = \{0 \mid 1\}$ gives the particular standard identity

$$\{\mathbb{N} \mid \omega/2^m : m \in \mathbb{N}\} = \omega^{1/2} = \sqrt{\omega}. \quad (9)$$

Formula (8) is a background construction, not an infinite summation convention introduced here [1, 2].

3 The broadcast rule

Definition 3.1 (Truncation maps). For a sign $\sigma \in \{+, -\}$ and an ordinal γ , set

$$T_\gamma^\sigma(s) = \begin{cases} s \restriction \delta, & \delta = \min\{\eta : \gamma \leq \eta < \ell(s), s(\eta) = \sigma\} \text{ exists,} \\ s, & \text{otherwise.} \end{cases} \quad (10)$$

For a finite set $F \subseteq I$, define

$$(T_{\gamma,F}^\sigma \mathbf{s})_i = \begin{cases} s_i, & i \in F, \\ T_\gamma^\sigma(s_i), & i \notin F. \end{cases} \quad (11)$$

The action is *legal* at $\mathbf{s}$ when some $j \notin F$ has $\gamma < \ell(s_j)$ and $s_j(\gamma) = \sigma$.

A legal plus action is a Left move and a legal minus action is a Right move. The associated recursive game is

$$G(\mathbf{s}) = \left\{ G(T_{\gamma,F}^+ \mathbf{s}) : \text{legal} \mid G(T_{\gamma,H}^- \mathbf{s}) : \text{legal} \right\}. \quad (12)$$

This reformulation of Lipparini's rule [5, Remark 7.6(3)] separates the globally defined truncation map from the requirement that a move have a witness at its exact threshold. That distinction will prevent a subtle mistake in the proof of number-valuedness.

For a fixed input, all reachable coordinates are prefixes of its original coordinates. The set of reachable states is consequently contained in a set-sized product of prefix sets, and the options at every state form sets. Section 4 establishes the well-foundedness needed to justify (12); no proper-class cut is used.

4 A descending ordinal rank

Fix an initial family and put

$$\Lambda = \sup_{i \in I} \ell(s_i), \quad \Omega = \omega^{\Lambda+1}. \quad (13)$$

These parameters remain fixed while considering followers of the initial family. For any current coordinate t_i , define its plus and minus heights by

$$p_i = \sup\{\eta + 1 : t_i(\eta) = +\}, \quad m_i = \sup\{\eta + 1 : t_i(\eta) = -\}, \quad (14)$$

with empty supremum 0. Both heights are at most Λ .

For any ordinal family $\mathbf{u} = (u_i)_{i \in I}$ bounded by Λ , set

$$\zeta(\mathbf{u}) = \min\{\xi : \{i : u_i > \xi\} \text{ is finite}\}, \quad (15)$$

$$h(\mathbf{u}) = \#_{i: u_i > \zeta(\mathbf{u})} u_i, \quad (16)$$

$$R(\mathbf{u}) = \Omega \cdot \zeta(\mathbf{u}) + h(\mathbf{u}). \quad (17)$$

The minimum exists since $\xi = \Lambda$ qualifies. The sum in (16) is finite. Moreover $h(\mathbf{u}) < \Omega$, since its summands are below the pure power Ω and finite natural sums preserve that bound. The form of ζ is inspired by the ordinal threshold invariant used in [5]; our rank tracks both polarities.

Lemma 4.1 (Monotonicity of the auxiliary rank). *If $v_i \leq u_i$ for all i , then $R(\mathbf{v}) \leq R(\mathbf{u})$. If additionally $\zeta(\mathbf{v}) = \zeta(\mathbf{u})$ and at least one coordinate above this common threshold strictly decreases, then $R(\mathbf{v}) < R(\mathbf{u})$.*

Proof. Coordinatewise decrease implies $\zeta(\mathbf{v}) \leq \zeta(\mathbf{u})$. If the inequality is strict, then

$$R(\mathbf{v}) < \Omega(\zeta(\mathbf{v}) + 1) \leq \Omega\zeta(\mathbf{u}) \leq R(\mathbf{u}).$$

If the thresholds agree, every new exceptional coordinate was already exceptional. Write both finite natural sums over the old exceptional index set, using zero for a coordinate that is no longer exceptional. The assertion follows from monotonicity, or strict monotonicity, of finite natural sum. $\square$

Theorem 4.2 (Well-foundedness and a birthday bound). *For the current family $\mathbf{t}$, let $\mathbf{p}, \mathbf{m}$ be as in (14), and define*

$$\mathcal{R}(\mathbf{t}) = R(\mathbf{p}) \# R(\mathbf{m}). \quad (18)$$

Every legal broadcast move strictly decreases $\mathcal{R}$. Consequently $G(\mathbf{s})$ is well-defined, no run is infinite, and its game-form birthday is at most $\mathcal{R}(\mathbf{s}) < \Omega^2$.

Proof. Every truncation replaces strings by prefixes, so both polarity-height families decrease coordinatewise. Both auxiliary ranks therefore weakly decrease by Lemma 4.1.

Consider a legal Left move at threshold γ , with witness $j \notin F$. After the move, every unprotected coordinate has plus height at most γ : there are no plus signs between γ and its cutting position, and all later signs are deleted. Thus $\zeta(\mathbf{p}') \leq \gamma$. The witness satisfies $p_j > \gamma$ and $p'_j \leq \gamma < p_j$.

If the plus threshold ζ decreases, its rank strictly decreases. Otherwise both thresholds agree and are at most γ . The witness was an exceptional coordinate and strictly decreased, so the second assertion of Lemma 4.1 again gives $R(\mathbf{p}') < R(\mathbf{p})$. The minus rank does not increase. Strict monotonicity of natural sum gives a strict decrease of (18). The Right case exchanges the two signs.

All reachable states form a set, and their ranks are ordinals. Transfinite recursion on this rank therefore defines the game. Induction also gives $b(G(\mathbf{t})) \leq \mathcal{R}(\mathbf{t})$: each follower has rank strictly smaller than the current rank, so its birthday plus one is at most the current rank.

Finally, $\Lambda + 1 < \Omega$ and $R(\mathbf{u}) < \Omega(\Lambda + 1) < \Omega^2$. The ordinal Ω^2 is again a pure power of ω , hence closed under finite natural sums of smaller ordinals. Thus $\mathcal{R}(\mathbf{t}) < \Omega^2$. $\square$

Remark 4.3. There need not be a uniform finite bound on the length of runs. Termination means that each individual run is finite, not that the game tree has finite height. The rank bound is deliberately generous and is not asserted to be optimal.

5 Commuting truncations force number-valuedness

Lemma 5.1 (Commutation). *For fixed signs, thresholds, and finite exemption sets, the truncation maps commute:*

$$T_{\gamma,F}^{\sigma} T_{\eta,H}^{\tau} \mathbf{s} = T_{\eta,H}^{\tau} T_{\gamma,F}^{\sigma} \mathbf{s}. \quad (19)$$

The maps need not both be legal at the states where they are applied.

Proof. Work in one coordinate. If either action exempts that coordinate, the claim is immediate. Otherwise let a and b be its two cutting positions computed in the original string, treating an absent cutting position as the full string length. Applying the two prefix operations in either order gives the prefix of length $\min(a, b)$. If the first cut erases the sign targeted by the second, the second operation becomes the identity; this still gives the same minimum. $\square$

Lemma 5.2 (A nonidentity action can be legalized). *If $T_{\gamma,F}^{\sigma} \mathbf{s} \neq \mathbf{s}$, its outcome is a legal move by the σ -player, possibly at a threshold larger than γ .*

Proof. Choose

$$\delta = \min\{\eta \geq \gamma : \exists i \notin F [\eta < \ell(s_i) \text{ and } s_i(\eta) = \sigma]\}.$$

This set is nonempty because the action is not the identity. There is a witness at δ , and no relevant unprotected sign occurs in $[\gamma, \delta)$. It follows coordinatewise that $T_{\gamma,F}^{\sigma} \mathbf{s} = T_{\delta,F}^{\sigma} \mathbf{s}$. $\square$

Lemma 5.3 (The earlier witness survives). *Suppose a Left action at γ and a Right action at η are both legal at $\mathbf{s}$. If $\gamma \leq \eta$, the original Left action is still legal after the Right action. If $\eta \leq \gamma$, the original Right action is still legal after the Left action.*

Proof. In the first case, take the chosen plus witness at position γ . A Right cut, unless the coordinate is protected, occurs at a minus position at least η . This position is strictly larger than γ : equality would require a sign to be both plus and minus. Thus the witness survives. The other case is symmetric. $\square$

Theorem 5.4 (Every broadcast game is a number). *For every set-indexed family of surreal numbers in their canonical sign expansions, $G(\mathbf{s})$ is a numerical game. Its surreal value $B(\mathbf{s})$ is therefore defined.*

Proof. Induct on the rank of Theorem 4.2. All options are numbers by induction. It remains to compare an arbitrary Left option and an arbitrary Right option. Write their states as

$$\mathbf{s}_L = T_{\gamma,F}^+ \mathbf{s}, \quad \mathbf{s}_R = T_{\eta,H}^- \mathbf{s},$$

and use Lemma 5.1 to define the common state

$$\mathbf{s}_* = T_{\eta,H}^- \mathbf{s}_L = T_{\gamma,F}^+ \mathbf{s}_R.$$

If $\gamma \leq \eta$, Lemma 5.3 makes $G(\mathbf{s}_*)$ a genuine Left option of $G(\mathbf{s}_R)$, so $B(\mathbf{s}_*) < B(\mathbf{s}_R)$. Starting instead from $\mathbf{s}_L$, the action leading to $\mathbf{s}_*$ is either the identity or, by Lemma 5.2, a genuine Right move at an appropriate threshold. Hence

$$B(\mathbf{s}_L) \leq B(\mathbf{s}_*) < B(\mathbf{s}_R).$$

If $\eta \leq \gamma$, the symmetric argument gives

$$\mathbf{B}(s_L) < \mathbf{B}(s_*) \leq \mathbf{B}(s_R).$$

Thus every left option is smaller than every right option. Since all options are numbers, the recursive criterion makes $G(s)$ a number. $\square$

Remark 5.5 (Why a tempting shortcut would be invalid). One cannot assume that both original thresholds remain legal after the other move. A sign may be erased. The proof needs the stronger distinction between a truncation *map* and a legal *move*: at least one witness survives, and the other action can be legalized or is the identity. Nor does the proof assume that $\mathbf{B}$ is coordinatewise monotone; that property will be disproved.

The canonical birthday of $\mathbf{B}(s)$ is no greater than the birthday of the numerical game form $G(s)$, so Theorem 4.2 also bounds the birthday of the resulting surreal number.

6 Basic identities and the ordinal restriction

Proposition 6.1 (Relabelling, zeros, and negation). *The operation $\mathbf{B}$ is invariant under bijections of the index set and under inserting or removing zero coordinates. Also*

$$\mathbf{B}((-s_i)_{i \in I}) = -\mathbf{B}((s_i)_{i \in I}). \quad (20)$$

Proof. A bijection carries finite exemption sets and legal witnesses to the corresponding ones, yielding the same recursive game. Empty strings supply no witness and are unchanged by every truncation, so zero coordinates can be ignored, including any of them lying in an exemption set. Negation reverses every sign, interchanges Left and Right actions, and commutes with prefix truncation. Induction on rank gives the usual game negation and hence (20). $\square$

Theorem 6.2 (Finite-support agreement). *If $s_i = 0$ outside a finite subset $J \subseteq I$, then*

$$\mathbf{B}(s) = \sum_{i \in J} s_i, \quad (21)$$

where the right side is ordinary finite surreal addition.

Proof. Induct on the rank of the family. Every follower still has finite support, so its value is the finite sum of its coordinates by induction.

Every ordinary single-coordinate move in the disjunctive sum is available as a broadcast move: exempt all the other nonzero coordinates. Conversely, a Left broadcast move replaces every changed coordinate by one of its left prefixes. Select its witnessing coordinate j . The ordinary move changing only j to the same prefix is a Left option of the ordinary sum and has value at least as large as the broadcast follower, because all other coordinate changes weakly lower the finite sum. The analogous Right comparison reverses the inequality. Thus the Left option sets are mutually cofinal and the Right option sets mutually coinital. Their numerical cuts have the same value. $\square$

Corollary 6.3 (Ordinal extension). *On countable families of ordinals, $\mathbf{B}$ equals the infinitary ordinal operation $\#^S$ characterized by Lipparini's Definition 7.1 and Theorem 7.3.*

Proof. For the all-plus sign string of length α_i , a legal plus action at γ replaces an unprotected coordinate by $\min(\alpha_i, \gamma)$. There are no Right moves. This is exactly the ordinal game, whose value is $\#^S$ by [5, Theorem 7.3]. $\square$

Remark 6.4 (Dependence on the index set). For any infinite set I , $B((1)_{i \in I}) = \omega$: a Left move retains an arbitrary finite number of ones, and there are no Right moves. Thus the set-indexed version is not a cardinal-valued counting operation. Its extension beyond sequences is a consequence of the same rules, not a proposed solution to every uncountable summation problem.

7 A finite-short-perturbation estimate

A useful comparison principle survives despite the later failure of coordinatewise monotonicity. It controls changes in a finite number of finite-birthday coordinates.

Theorem 7.1 (Finite-short-perturbation bound). *Suppose $\mathbf{s}$ and $\mathbf{t}$ have the same index set and coincide outside a finite set E . Suppose that s_i and t_i are short for $i \in E$. Define*

$$N = \sum_{i \in E} (\ell(s_i) + \ell(t_i)) \in \mathbb{N}. \quad (22)$$

Then

$$|B(\mathbf{s}) - B(\mathbf{t})| \leq N. \quad (23)$$

Proof. Use a common ordinal bound Λ for both families and the same rank parameters as in Section 4. Prove $B(\mathbf{s}) \leq B(\mathbf{t}) + N$ by induction on $\mathcal{R}(\mathbf{s}) \# \mathcal{R}(\mathbf{t})$, keeping the finite set E fixed.

Apply a truncation map, with the same sign, threshold, and exemption set, to both families, giving $\mathbf{s}'$ and $\mathbf{t}'$. These still agree outside E , and the new budget N' is at most N . By Lemma 5.2, each action is either a legal move or the identity. If one side changes and the other does not, some coordinate in E must have lost at least one sign: outside E the inputs and actions agree. Since all lengths in E are finite,

$$\text{one action proper and the other idle} \implies N' \leq N - 1. \quad (24)$$

Whenever at least one side changes, the rank of the pair strictly decreases, so the induction hypothesis applies to the resulting pair.

For a Left move $\mathbf{s} \rightarrow \mathbf{s}'$, match the action on $\mathbf{t}$. If the matched action is a proper Left move, then

$$B(\mathbf{s}') \leq B(\mathbf{t}') + N' \leq B(\mathbf{t}') + N < B(\mathbf{t}) + N.$$

If it is idle, (24) instead gives

$$B(\mathbf{s}') \leq B(\mathbf{t}) + N' \leq B(\mathbf{t}) + N - 1 < B(\mathbf{t}) + N.$$

Thus every Left option of $G(\mathbf{s})$ lies below $B(\mathbf{t}) + N$.

For a Right move $\mathbf{t} \rightarrow \mathbf{t}'$, match it on $\mathbf{s}$. If the matched action is a proper Right move, then

$$B(\mathbf{s}) < B(\mathbf{s}') \leq B(\mathbf{t}') + N' \leq B(\mathbf{t}') + N.$$

If it is idle, induction and (24) give

$$B(\mathbf{s}) \leq B(\mathbf{t}') + N' \leq B(\mathbf{t}') + N - 1 < B(\mathbf{t}') + N.$$

Use the game form $G(\mathbf{t}) + N$, with N its nonnegative integer form. Its Right options are exactly $G(\mathbf{t})^R + N$, since N has no Right options. The two groups of inequalities just proved are precisely the comparison test (7) for $B(\mathbf{s}) \leq B(\mathbf{t}) + N$. Interchanging the two families proves the reverse bound. $\square$

Corollary 7.2. *If a family of short numbers agrees outside a finite set with a constant short number r , then its broadcast value differs from $B(r, r, \dots)$ by a finite surreal number.*

The corollary is not an infinitesimal estimate, an integer-error formula, or an ℓ^1 -continuity theorem. It only says that the error is bounded in absolute value by a finite integer. The distinction will be enough to identify the leading scale of all Right options in the main example.

8 Constant dyadic sequences and finite defects

For $m \in \mathbb{N}$, write

$$r_m = 2^{-m}, \quad X_m = \omega r_m. \quad (25)$$

All products and differences in this section are in the surreal field.

8.1 Two standard cut identities

Lemma 8.1. *The following cuts hold:*

$$X_0 = \{ \mathbb{N} \mid \}, \quad (26)$$

$$X_m = \{ \mathbb{N} \mid X_{m-1} - n : n \in \mathbb{N} \} \quad (m \geq 1), \quad (27)$$

$$X_h - k = \{ \mathbb{N} \mid X_h - (k-1) \} \quad (h \geq 0, k \geq 1). \quad (28)$$

Proof. The first identity is the definition of ω . Since $r_m = \{ 0 \mid r_{m-1} \}$ for $m \geq 1$, the Conway product with $\omega = \{ \mathbb{N} \mid \}$ gives

$$X_m = \{ nr_m : n \in \mathbb{N} \mid X_{m-1} - nr_m : n \in \mathbb{N} \}.$$

The left side is mutually cofinal with $\mathbb{N}$; replacing nr_m by n on the right is also a mutually cointial replacement. This gives (27).

For (28), use the finite-sum form of $X_h + (-k)$, where $-k = \{ \mid -(k-1) \}$. Its left options are real numbers cofinal with $\mathbb{N}$. One right option is $X_h - (k-1)$. If $h \geq 1$, the other right options have the form $X_{h-1} - n - k = 2X_h - n - k$, all strictly above $X_h - (k-1)$, since X_h exceeds every real number. They can be discarded. For $h = 0$ there are no such additional right options. $\square$

Theorem 8.2 (Constant dyadic values and finite defects). *For every $m \in \mathbb{N}$,*

$$B(r_m, r_m, \dots) = X_m. \quad (29)$$

For $m \geq 1$, the family with infinitely many r_{m-1} and exactly k entries equal to r_m has value

$$H_{m,k} = X_{m-1} - k. \quad (30)$$

The positions of the finitely many exceptional entries do not matter.

Proof. We use induction on m . The constant sequence of ones has only Left moves, which give arbitrary finite collections of ones. Its cut is $\{\mathbb{N} \mid \} = \omega$, proving (29) for $m = 0$.

Assume the constant formula for all smaller indices. For this fixed $m \geq 1$, first prove (30) by induction on k . Its base case is the already established constant value X_{m-1} .

The sign expansion of r_j is a plus followed by j minuses. Hence the only Left threshold of the family defining $H_{m,k}$ is 0. Such a move leaves finitely many positive dyadic entries and has their ordinary finite sum as its value, by Theorem 6.2. Since infinitely many r_{m-1} are available, these left values are unbounded finite reals and are mutually cofinal with $\mathbb{N}$.

A Right move at threshold m can act only on the k exceptional entries. It changes each unprotected r_m into r_{m-1} , producing precisely values $H_{m,j}$ with $0 \leq j < k$. Every such j can be obtained by protecting the appropriate exceptional entries. By the induction on k , the smallest such Right value is $X_{m-1} - (k - 1)$.

A Right move at a threshold $q < m$, necessarily $q \geq 1$, produces a family that is constantly r_{q-1} outside finitely many short exceptions. The perturbation bound and the outer induction show that its value is $X_{q-1} + \varepsilon$ for a finite surreal ε . Here $q - 1 \leq m - 2$, so $X_{q-1} \geq 2X_{m-1}$. Such a value is strictly greater than X_{m-1} plus any prescribed finite integer, and is therefore greater than $X_{m-1} - (k - 1)$. These other Right options impose no stronger constraint. The cut for $H_{m,k}$ is consequently

$$\{\mathbb{N} \mid X_{m-1} - (k - 1)\} = X_{m-1} - k$$

by Lemma 8.1. This finishes the induction on k .

Now consider the constant r_m family. Its Left values are again finite reals cofinal with $\mathbb{N}$. A Right move at threshold m , with exactly k entries protected, gives $H_{m,k} = X_{m-1} - k$, for every $k \in \mathbb{N}$. Right moves at $q < m$ are larger by the same finite-perturbation argument. Hence its cut is

$$\{\mathbb{N} \mid X_{m-1} - k : k \in \mathbb{N}\} = X_m.$$

This completes the outer induction. □

Corollary 8.3 (Cofinite dyadic leading scale). *If $\mathbf{t}$ is constantly r_m outside finitely many short exceptions, then there is an $N \in \mathbb{N}$ such that*

$$X_m - N \leq \mathbf{B}(\mathbf{t}) \leq X_m + N. \quad (31)$$

In particular $\mathbf{B}(\mathbf{t})$ exceeds every real number.

Proof. Combine Theorems 7.1 and 8.2. A positive real multiple of ω , minus a finite number, is still larger than every real. □

9 An exact classification of convergent dyadic series

Fix $d \geq 2$ and abbreviate $\mathbf{s}^{E,d}$ from (2) to $\mathbf{s}^E$ when the shift is understood. Let

$$a_n = 2^{-(n+d)}, \quad b_n = \frac{3}{2}a_n.$$

Their sign expansions are

$$a_n = + -^{n+d}, \quad b_n = + -^{n+d} +. \quad (32)$$

The second plus in b_n is at position $n + d + 1$, with positions counted from zero. Its deletion changes b_n exactly to a_n .

9.1 A complete description of the relevant options

Lemma 9.1 (Left options). *For any $E \subseteq \mathbb{N}$, the Left moves from s^E have the following form. A threshold-zero move has finite support and value strictly less than 1. A positive-threshold move produces $s^{E'}$ for a finite subset $E' \subseteq E$. If E is finite and nonempty, $|E'| < |E|$, and a move with $|E'| = |E| - 1$ is available. If E is infinite, every finite value of $|E'|$ is attainable.*

Proof. At threshold zero every unprotected positive string is replaced by the empty string. Only finitely many entries remain, and their sum is less than the total real sum, which is at most $3 \cdot 2^{-d} < 1$.

A positive Left threshold must be $\gamma = n + d + 1$ for a second plus at some $n \in E$. Coordinates of the base type a_j have no plus sign at or beyond γ and are unchanged. In an upgraded coordinate b_j , the extra plus is removed exactly when it occurs at or after γ and the coordinate is unprotected. Only finitely many second pluses occur before γ , and only finitely many indices are protected. Thus finitely many upgrades survive. The witnessing upgrade is removed, so for finite E the number strictly falls. Exempting all other upgrades permits exactly one removal.

If E is infinite, choose the threshold of its earliest upgrade. Exempt any k other upgraded indices. All unprotected upgrades are removed, and precisely k remain. This realizes every $k \in \mathbb{N}$. $\square$

Lemma 9.2 (Right options). *Every Right option from s^E is constantly $r_{\gamma-1}$ outside finitely many short exceptions, where $\gamma \geq 1$ is its threshold. Consequently its value is $X_{\gamma-1} + \varepsilon$, with ε finite, and is larger than every real number. Every integer threshold $\gamma \geq 1$ is legal.*

Proof. All minus signs occur at positive finite positions. For a fixed $\gamma \geq 1$, all sufficiently large indices satisfy $n + d \geq \gamma$. Both a_n and b_n have a minus at position γ , and truncation immediately before that position gives

$$+ - \gamma^{-1} = r_{\gamma-1}.$$

Only finitely many earlier coordinates and the finite exemption set can differ from this tail. All exceptional strings are short. Apply Corollary 8.3. Since arbitrarily long initial minus blocks are present, a witness exists at every positive finite threshold. $\square$

9.2 The finite-upgrade case

Theorem 9.3. *If E is finite, then $B(s^E) = |E| + 1$.*

Proof. Induct on $k = |E|$. For $k = 0$, all Left options are finite-support values less than 1, including 0, and all Right options exceed every real number. The simplest number between the two sides is 1: it fits the cut, whereas its only canonical predecessor 0 does not. Thus the value is 1.

For $k > 0$, Lemma 9.1 and induction show that a positive Left threshold gives a value $j + 1$ with $j < k$. Such values are at most k , and k itself is attained. Threshold-zero values remain below 1. Every Right option is infinite by Lemma 9.2. The simplest number larger than these Left values and smaller than these Right values is $k + 1$: it fits, and every canonical predecessor of the integer $k + 1$ is at most k and fails. $\square$

9.3 The infinitely-many-upgrades case

Theorem 9.4. *If E is infinite, then $B(s^E) = \sqrt{\omega}$.*

Proof. By Lemma 9.1 and Theorem 9.3, all Left options are finite numbers and include every positive integer. Thus the Left option set is mutually cofinal with $\mathbb{N}$.

Let $\mathcal{Q}$ be the set of Right option values. We show that it is mutually cointial with $\{X_m : m \in \mathbb{N}\}$.

First take $q \in \mathcal{Q}$. By Lemma 9.2, for some $\gamma \geq 1$ and some finite integer N ,

$$q \geq X_{\gamma-1} - N > X_\gamma.$$

The last inequality holds because $X_{\gamma-1} - X_\gamma = X_\gamma$ is infinite. So some X_m lies below every given Right value.

Conversely, fix $m \in \mathbb{N}$. Choose a Right move at threshold $\gamma = m + 2$ with empty exemption set; it is legal by Lemma 9.2. Its value q_m satisfies $q_m \leq X_{m+1} + N_m$ for some finite N_m , and therefore

$$q_m < X_m,$$

since $X_m - X_{m+1} = X_{m+1}$ exceeds every finite integer. Thus some Right value lies below each prescribed X_m .

Replacing both option sets by these mutually cofinal and cointial sets gives

$$\mathbf{B}(\mathbf{s}^E) = \{\mathbb{N} \mid X_m : m \in \mathbb{N}\}.$$

By the standard omega-map cut (9), this is $\omega^{1/2} = \sqrt{\omega}$. $\square$

Remark 9.5 (Sparsity does not affect the result). The proof uses only that E is infinite, not any positive density or bounded-gap assumption. For example, the squares, powers of two, or any arbitrarily sparse infinite set give the same output. For finite E , the output depends only on its cardinality, not on the locations or total real mass of the upgrades.

10 Consequences: monotonicity, scaling, and limits

Lemma 10.1 (Pure geometric tails). *For every integer $e \geq 1$,*

$$\mathbf{B}((2^{-(n+e)})_{n < \omega}) = 1. \quad (33)$$

Proof. The only Left threshold is zero. Its finite-support values are strictly below the total real sum $2^{1-e} \leq 1$, and include 0. A Right move at any positive threshold produces a constant positive dyadic tail with finitely many short exceptions, hence has infinite value by Corollary 8.3. The simplest number fitting this cut is 1. $\square$

Corollary 10.2 (Failure of weak monotonicity). *There are strictly positive absolutely convergent dyadic sequences $\mathbf{s}$ and $\mathbf{t}$ such that $s_n < t_n$ for every n but $\mathbf{B}(\mathbf{s}) > \mathbf{B}(\mathbf{t})$. The reversal can be an arbitrarily large integer or an infinite surreal.*

Proof. Set $\mathbf{s} = \mathbf{s}^{E,d}$ and $t_n = 2^{-(n+d-1)}$. Then $s_n \leq \frac{3}{2} 2^{-(n+d)} < 2 2^{-(n+d)} = t_n$. By Lemma 10.1, $\mathbf{B}(\mathbf{t}) = 1$. For nonempty finite E , the other value is $|E| + 1$; for infinite E , it is $\sqrt{\omega}$. $\square$

In the particularly simple all-upgraded case with $d = 2$, the reversal reads

$$\mathbf{B}(3/8, 3/16, 3/32, \dots) = \sqrt{\omega}, \quad (34)$$

$$\mathbf{B}(1/2, 1/4, 1/8, \dots) = 1. \quad (35)$$

Every term on the first line is strictly smaller than the corresponding term on the second. Their classical sums are $3/4$ and 1 respectively.

Corollary 10.3 (Failure of finite-scaling compatibility). *The operation is not homogeneous under multiplication by $3/2$, even on positive convergent dyadic sequences. It also need not satisfy $B(s + s) = 2B(s)$.*

Proof. For $a_n = 2^{-(n+2)}$, the classification gives $B(a) = 1$, while $B((3/2)a) = \sqrt{\omega} \neq 3/2$. Also $B(2a) = 1$ by (33), not 2. $\square$

10.1 A numerical table with exact surreal outputs

For $d = 2$, write $E_k = \{0, 1, \dots, k-1\}$, including $E_0 = \emptyset$. The ordinary sum is

$$\sum_{n=0}^{\infty} s_n^{E,2} = \frac{1}{2} + \sum_{n \in E} 2^{-(n+3)}, \quad \sum_{n=0}^{\infty} s_n^{E_k,2} = \frac{3}{4} - 2^{-(k+2)}. \quad (36)$$

The following values are therefore exact, but the two columns describe different operations.

Upgrade set or sequence	Classical real sum	Broadcast value
$E = \emptyset$	$1/2$	1
$E = \{0\}$	$5/8$	2
$E = \{0, 1\}$	$11/16$	3
$E = \{0, 1, 2\}$	$23/32$	4
$E = \mathbb{N}$	$3/4$	$\sqrt{\omega}$
$(1/2, 1/4, 1/8, \dots)$	1	1

10.2 Why finite truncations cannot establish the infinite answer

Let $s^{[N]}$ denote the family retaining only its first N coordinates and replacing the rest by zero. The finite-support theorem gives

$$B(s^{[N]}) = \sum_{n=0}^{N-1} s_n.$$

For the all-upgraded sequence with $d = 2$, this is

$$B(s^{[N]}) = \frac{3}{4}(1 - 2^{-N}) < \frac{3}{4}, \quad (37)$$

whereas $B(s) = \sqrt{\omega}$. Finite game evaluation and subsequent real limiting would therefore compute the wrong operation.

There is no contradiction. An infinite broadcast move truncates infinitely many coordinates at once while allowing only finitely many exemptions. Its game tree is not a numerical limit of the values of finite-support truncations. The proof of the infinite value relies on its full option sets, especially their cofinal and coinital structure.

10.3 Small real mass does not imply a small output

For every real $\varepsilon > 0$, choose d so large that $3 \cdot 2^{-d} < \varepsilon$. The all-upgraded family then has ℓ^1 norm below ε , but its broadcast value remains $\sqrt{\omega}$. Thus no bound by a real-valued function of the ordinary ℓ^1 norm can hold uniformly for these inputs.

Even a perturbation at a single sufficiently late index has arbitrarily small real mass but changes the value of the base family from 1 to 2. The finite-short-perturbation theorem is consistent with this behavior: it uses sign lengths, not numerical size, as its budget.

11 Computational checks and reproducibility

The archive includes a standard-library Python implementation using exact rational arithmetic. It recursively constructs the *actual finite game* for finite families of finite canonical sign strings, checks separation of Left and Right option values, and chooses the simplest dyadic in the resulting cut. It does not insert the finite-support theorem as the evaluation rule.

Running `python code/verify.py` produced the following test counts:

Check	Count
Single-string commutation, sign length at most 8	135,156
Finite two-coordinate game values, lengths at most 4	961
Finite three-coordinate game values, lengths at most 3	3,375
Legal Left–Right diamonds, including exemption sets	3,378
Input pairs checked for negation, reversal, and zero insertion	225
Dyadic sign formulas and positive-constant-tail truncations	14,560

All checks passed. The recursive evaluator memoized 4,336 finite states. The exact report, including the scope limitation, is saved as `verification/test_report.json`. Runtime depends on the environment.

The diamond tests explicitly check both details used in the proof: the earlier witness survives, and the other action is either idle or a legal move after a possible threshold adjustment. Merely testing commutation would miss the second issue.

The file `verification/partial_sums.csv` contains exact finite partial sums for the base and all-upgraded $d = 2$ families. Its columns naming the infinite values are labeled as values *from the theorem*; they are not numerical outputs of an infinite-surreal evaluator.

What the checks establish. They test the implementation and provide finite consistency checks for the local lemmas and finite-support theorem. They do not prove well-foundedness for arbitrary ordinal-length strings, the universal number theorem, the finite-perturbation estimate for long tails, or the value $\sqrt{\omega}$. Those statements are supported by the written proofs, not by extrapolation from finite experiments. No proof-assistant formalization is included.

12 Research assessment and remaining questions

12.1 Why this attack was tractable

The rule is governed by canonical sign positions rather than arbitrary comparisons of surreal magnitudes. Prefix truncations have a rigid algebra: they commute, and an earlier opposite-sign witness survives. That gives a local mechanism forcing the global game to be a number.

The same sign-based mechanism creates the counterexample. A tiny dyadic increase from a_n to b_n appends one plus sign at a late position. A single broadcast move can remove all but finitely many such signs. Thus the *finite versus infinite* combinatorics of the appended signs matter far more than the real sizes of the perturbations. The surviving finite cases supply all positive integers as Left options; constant dyadic tails control the Right options. The final answer is forced by the cut between these two scales.

This explains why the operation can be a valid numerical extension of an ordinal game while being unsuitable as an order-preserving completion of ordinary real summation. Number-valuedness is a game-theoretic property; coordinatewise monotonicity is an additional assertion about its dependence on input values. One does not imply the other.

12.2 Precisely what remains open in this manuscript

An alternative rule might preserve weak coordinatewise monotonicity while extending the ordinal operation. Nothing proved here excludes that possibility. One would have to state the desired compatibility axioms explicitly; the particular broadcast rule already fails both classical convergence agreement and even elementary finite-scaling identities.

Several narrower directions also remain untreated. The exact value of $B(r, r, \dots)$ for a general positive surreal r is not established here; Theorem 8.2 covers the dyadics 2^{-m} , not all numbers. Our finite-short-perturbation estimate does not cover arbitrary finite changes in long surreal coordinates. The ordinal birthday bound is not sharp. And the classification of positive dyadic sequences with more than one extra sign per coordinate may exhibit additional scales, but no formula for them is claimed.

A useful continuation would be to characterize a substantial class of inputs on which the broadcast operation is monotone, or to determine how a controlled hierarchy of late sign patterns changes its output cut. These are concrete questions suggested by the proofs rather than claims that the broad extension problem has been settled.

12.3 Provenance and priority limits

The original problem and the exact proposed rule were checked directly against the extended source, including the rendered page containing the sign rule. The arXiv record identifies v2 as an extended version of the published article. Related game-series papers were also identified, but this was not an exhaustive bibliographic review. In particular, the present document does not assert that no independent proof or overlapping result exists elsewhere.

The construction belongs to Lipparini. The standard facts about surreal cuts, comparison, finite arithmetic, and the omega-map belong to the established literature. The explicit arguments and classification in this manuscript are submitted as mathematical research claims to be checked, not presented as results already certified by those sources.

13 Conclusion

The proposed sign-truncation rule is more structured than a general infinite game: it always produces a surreal number. A well-founded ordinal rank makes the recursion precise, and a commuting-move argument separates every pair of opposite options. Finite-support agreement and an explicit perturbation bound then permit exact transfinite calculations.

The decisive calculation is

$$\mathbf{B} \left(\left(2^{-(n+d)} \left(1 + \frac{1}{2} \mathbf{1}_E(n) \right) \right)_{n < \omega} \right) = \begin{cases} |E| + 1, & E \text{ finite,} \\ \sqrt{\omega}, & E \text{ infinite.} \end{cases}$$

All these inputs have small convergent positive real sums. A strictly larger geometric input can have broadcast value 1. This is both positive progress on the number-valued extension question and an exact obstruction to using this particular candidate as a weakly monotone summation theory.

A Proof dependency audit

The main arguments have the following noncircular dependency structure.

Result	Inputs actually used
Descending rank	Prefix lengths; ordinal well-ordering; finite natural sum
Number-valuedness	Rank induction; commutation; legalizing nonidle maps; surviving earlier witnesses; numerical option comparison
Finite-support agreement	Number-valuedness; rank induction; monotonicity of <i>finite ordinary</i> surreal sums
Finite-short perturbation	Number-valuedness; pair-rank induction; finite birthday budget; game comparison
Constant dyadic values	Finite-support agreement; perturbation; ordinary product cuts; nested finite induction
Finite-upgrade classification	Option descriptions; constant-tail estimates; simplicity of natural numbers
Infinite-upgrade classification	Finite-upgrade values; constant-tail estimates; mutual coinitality; the standard omega-map cut
Failure of monotonicity	Exact classification; separate pure-power calculation

In particular, no step uses coordinatewise monotonicity of $\mathbf{B}$. The only monotonicity of numerical addition used in the finite-support proof is the standard monotonicity of an already defined *finite* sum.

B A small worked sign example

For $d = 2$, the first few base and upgraded strings are

n	a_n	signs of a_n	b_n	signs of b_n
0	1/4	+ - -	3/8	+ - - +
1	1/8	+ - - -	3/16	+ - - - +
2	1/16	+ - - - -	3/32	+ - - - - +

Consider the infinite all-upgraded family. A Left move at position 3 with no exemptions deletes every extra plus: all their positions are at least 3. The result is the base family, whose value is 1. Protecting any k upgraded coordinates other than the selected witness leaves exactly k upgrades and produces value $k + 1$.

A Right move at position 2 with no exemptions changes every entry to the prefix $+- = 1/2$, giving the constant family of value $\omega/2$. A Right move at a much later position leaves finitely many early exceptions and produces a much smaller positive dyadic constant tail. Its value is a smaller positive dyadic multiple of ω , up to a finite error. These observations are the concrete form of the two option-set arguments in Theorem 9.4.

C Archive contents and build instructions

The archive is self-contained apart from standard Python and a LaTeX installation. It includes the source `article.tex`, compiled `article.pdf`, bibliography `references.bib`, a `build.sh` script, a `README.md`, two Python modules under `code/`, and exact verification artifacts under `verification/`. No third-party paper or font file is redistributed.

Run the finite checks with

```
python code/verify.py
```

Build the paper with

```
sh build.sh
```

The latter runs `pdflatex`, `bibtex` (or `bibtex8`), and two further `pdflatex` passes. The bibliography records source identifiers and links; links were checked during preparation on September 20, 2026. A rebuild of the PDF is not a verification of the mathematical proofs.

References

- [1] John H. Conway. *On Numbers and Games*. A K Peters, Natick, MA, second edition, 2001.
- [2] Harry Gonshor. *An Introduction to the Theory of Surreal Numbers*, volume 110 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, 1986.
- [3] Paolo Lipparini. Series of combinatorial games. [arXiv:2406.02453](#), 2024. A different infinitary game construction; listed for context.
- [4] Paolo Lipparini. Infinite sums of combinatorial games (Dadaist games). [arXiv:2505.00614](#), 2025. A different infinitary game construction; listed for context.
- [5] Paolo Lipparini. Monotone infinitary operations on ordinals (extended version). [arXiv:2505.00424v2](#), 2026. Version 2, April 30, 2026. Problem 7.5 and Remark 7.6(3), pp. 36–37. Related published article: A Monotone Infinitary Operation on Ordinals, *Mathematical Logic Quarterly* 72 (2026), doi:10.1002/malq.70019.
- [6] Dierk Schleicher and Michael Stoll. An introduction to Conway’s games and numbers. [arXiv:math/0410026v2](#), 2005. Version 2, September 30, 2005.